Unit 3 e-Portfolio Activity: Correlation and Regression

Introduction

Four notebooks were rerun, one value altered per run from a restarted kernel, to see how single changes shape results.

What changed and what happened

Correlation. More noise cut r but covariance barely moved (Table 1, Figure 1): independent noise inflates only var(y), and r, unlike covariance, is standardised (University of Essex Online, 2026).

Simple regression. Adding 20 to every y moved only the intercept. Moving one point to (17, 150) reversed the slope (Figure 2), because edge points carry leverage (James *et al.*, 2013, pp. 97-99).

Multiple regression. Both supplied cases outweigh the heaviest car. Adding 50 g/km to the lightest car flipped a weight coefficient barely larger than its standard error (Table 1). Correlated predictors (0.754) widen such errors (James *et al.*, 2013, pp. 99-102).

Polynomial regression. Degree 6 fitted best, yet the cubic predicted best of the four under leave-one-out validation (Table 1, Figure 3); extra terms chase noise (Bishop, 2006, pp. 6-11). Selecting the degree on that noisy score leaves no independent test (Géron, 2025, ch. 1). Raising one interior point cut the cubic's training R^2 to 0.537 yet moved its prediction at 17 by one unit (Figure 4): interior points degrade fit, edge points redirect the model.

Exercise (n)	Baseline	Change	Result
Correlation (1,000)	r = 0.888; covariance 389.75	Noise SD 10 to 40	r = 0.446; covariance 403.02
		y + 500	r and covariance unchanged
		y × 10	r unchanged; covariance × 10
Simple regression (13)	Slope -1.75; R^2 0.58	(17, 86) replaced by (17, 150)	Slope +1.12; R^2 0.06
Multiple regression (36)	107.21 g/km at 2,300 kg, 1,300 cm^3; training R^2 0.38, leave-one-out R^2 0.22	Supplied second case: weight 3,300 kg	114.76 g/km; both weights above the observed 790 to 1,746 kg

		Author's case: volume 2,300 cm³	115.01 g/km
		Author's case: lightest car's CO_2 + 50 g/km, refit	Weight coefficient +0.0076 to -0.0046 g/km per kg (standard error 0.0064, t = 1.17)
Polynomial (18)	Cubic: training R^2 0.943; leave-one-out RMSE 4.23	Degree 6	Training R^2 0.980; leave-one-out RMSE 6.38
		Speed at hour 10: 70 to 110, cubic refit	Training R^2 0.537; prediction at 17 from 88.87 to 89.98

Table 1: Baselines, changes and results from the runs described above, each from a restarted kernel with all cells executed. Data: the four supplied teaching sets (a simulated pair, seed 1, n = 1,000; 13 points; 36 cars; 18 hourly speeds)

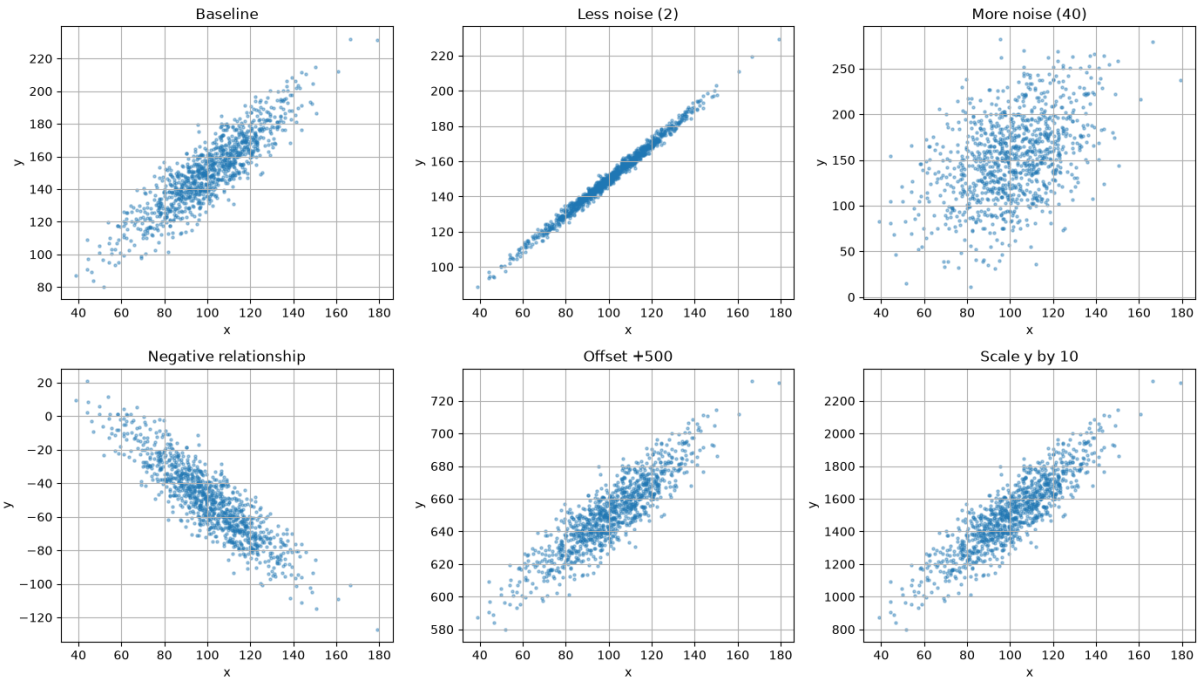


Figure 1: The six correlation experiments; the top row varies the noise, the bottom row reverses, shifts and rescales y

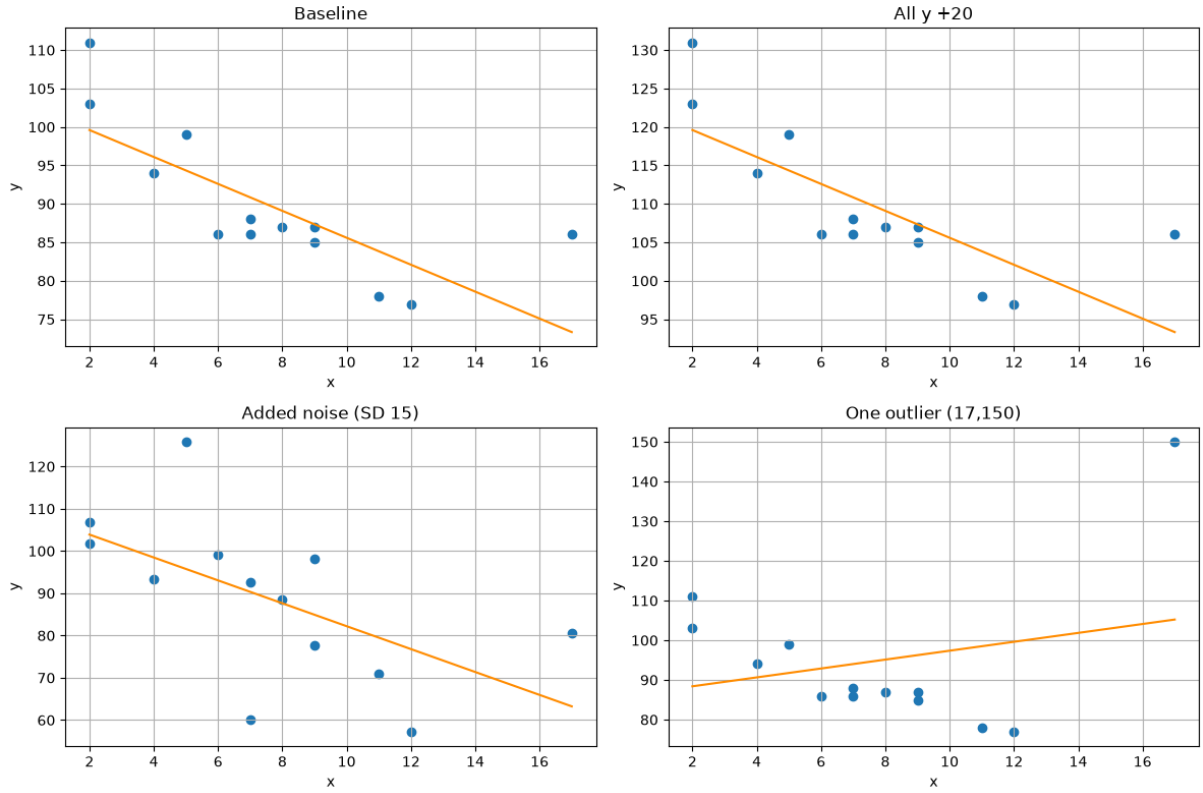


Figure 2: Simple regression under a shift, added noise and one outlier

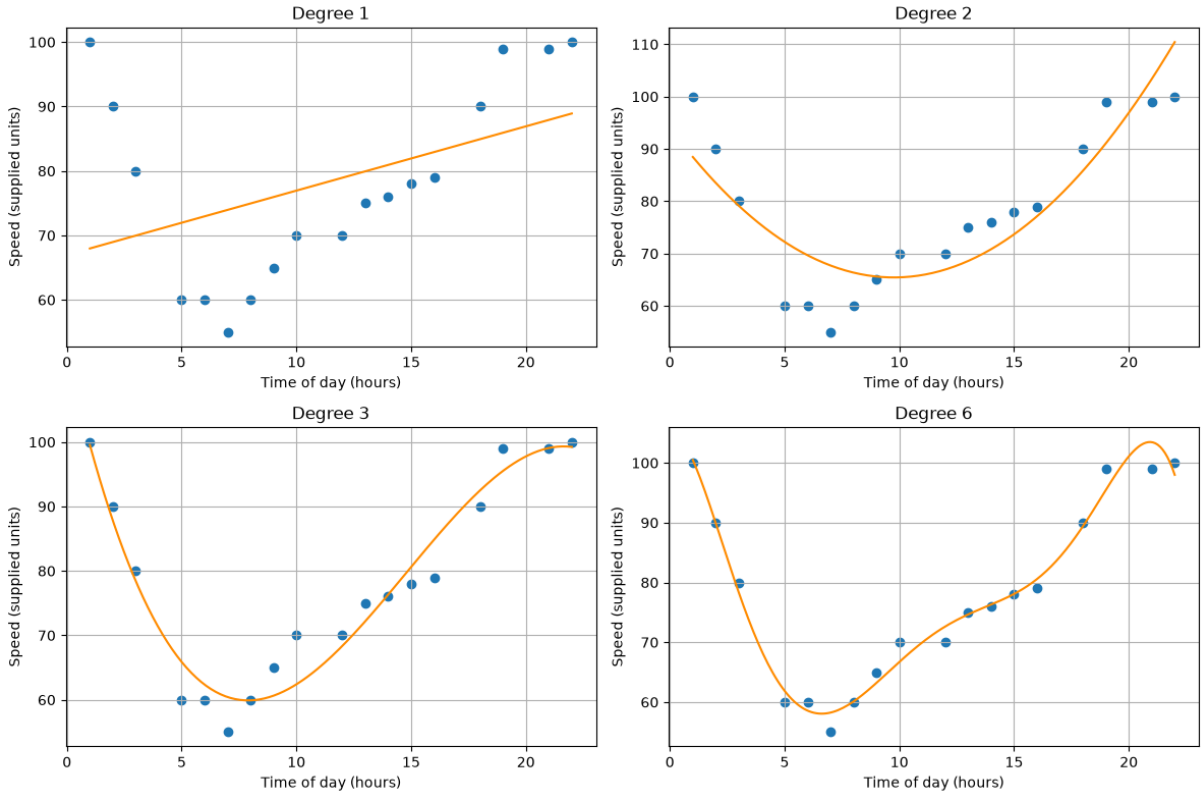


Figure 3: Polynomial fits of degree 1, 2, 3 and 6 to the same 18 points

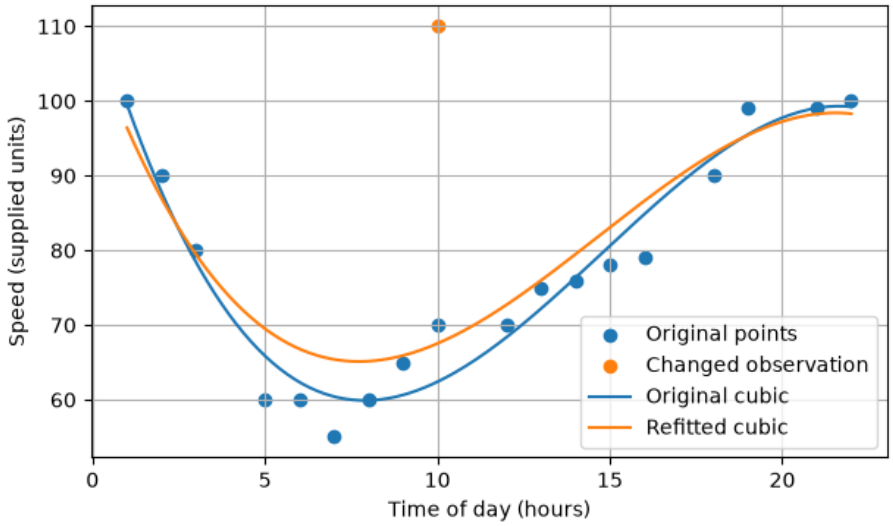


Figure 4: The cubic refitted after the speed at hour 10 is changed from 70 to 110

Implications for practice

Two notebooks failed as supplied (the name stat undefined, cars.csv missing) yet held saved outputs; professionally, no output counts until a clean rerun. Ethically, the coefficient is a predictive effect, not a cause (Chernozhukov *et al.*, 2024, ch. 1). Legally, the substituted cars data are W3Schools' example, reusable for non-profit teaching but 'for testing purposes only' (W3Schools, no date a, b). Without personal data, they cannot test group fairness. Socially, an extrapolation could steer decisions.

Conclusion

The sign reversals came from single edge points: one outlier, one CO_2 value. A model this sensitive should state its observed range and validation error, not its fit, before use.

References

Bishop, C.M. (2006) *Pattern Recognition and Machine Learning*. New York: Springer.

Chernozhukov, V. *et al.* (2024) *Applied Causal Inference Powered by ML and AI*. Version 0.1.1. Available at: https://causalml-book.org/ (Accessed: 22 October 2026).

Géron, A. (2025) *Hands-On Machine Learning with Scikit-Learn and PyTorch*. Santa Rosa: O'Reilly.

James, G. *et al.* (2013) *An Introduction to Statistical Learning: with Applications in R*. New York: Springer. Available at: https://www.statlearning.com/s/ISLR-Seventh-Printing.pdf (Accessed: 22 October 2026).

University of Essex Online (2026) 'Unit 3 Lecturecast: Correlation and Regression' [Lecturecast]. *ML_PCOM7E: Machine Learning*. University of Essex Online. Available at: https://www.my-course.co.uk/pluginfile.php/1659683/mod_scorm/content/2/scormcontent/index.html (Accessed: 22 October 2026).

W3Schools (no date a) *Python Machine Learning Multiple Regression: data.csv* [Dataset]. Available at: https://www.w3schools.com/python/python_ml_multiple_regression.asp (Accessed: 22 October 2026).

W3Schools (no date b) *About Terms of Service*. Available at: https://www.w3schools.com/about/about_copyright.asp (Accessed: 22 October 2026).

AI Acknowledgement

Codex was used to assist in identifying appropriate code solutions and to proofread and refine the code I produced. Claude and Grammarly supported proofreading, language refinement, and formatting of my written text. All final decisions, analyses, interpretations, conclusions, and recommendations are my own.